

Examiner  
only

6. A group of scientists studied the sand dune ecosystem at Ynyslas, Ceredigion. They cleared a 10 m by 10 m section of land in the dunes to expose the soil. The position of this cleared area is shown by the arrow in the photograph below.



land cleared here  
(shown prior  
to clearance)

Each year, they randomly selected twenty 1 m<sup>2</sup> areas in the study area and identified the different species present. They noted how many plants of each species were present.

- (a) (i) Describe how and explain why the sites are selected at random. [3]

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The table below shows the mean number of each species in a 1m<sup>2</sup> area after five years.

| Species            | Mean number of plants per m <sup>2</sup> |
|--------------------|------------------------------------------|
| Ragwort            | 3                                        |
| Birdsfoot trefoil  | 6                                        |
| Hairy hawkbit      | 2                                        |
| Eyebright          | 9                                        |
| Rosebay willowherb | 4                                        |
| Dandelion          | 2                                        |
| Mouse ear          | 4                                        |
| Restharrow         | 7                                        |
| Kidney vetch       | 1                                        |
| Groundsel          | 18                                       |
| Grass species 1    | 4                                        |
| Grass species 2    | 5                                        |

- (ii) In the first year the scientists only found four different species in the area. Identify the type of succession that has taken place in the five years of the study and explain why the number of plant species increased. [3]

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- (iii) Suggest what would happen to the number of animal species in this area. Explain your answer. [2]

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(b) Birdsfoot trefoil and restharrow are legumes, therefore they have root nodules that contain the bacterium *Rhizobium*. The scientists observed that they were two of the first plants to arrive in areas with nutrient poor soils.

(i) Explain why they are able to survive in soils with low nitrogen levels. [3]

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(ii) *Rhizobium* contain the enzyme nitrogenase which is responsible for fixing atmospheric nitrogen. This enzyme is inhibited if oxygen levels are high. The root nodules of legumes contain a form of haemoglobin called leghaemoglobin. This has a very high affinity for oxygen.

Suggest why leghaemoglobin is present in the root nodules of restharrow and birdsfoot trefoil. [3]

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(c) Close to Ynyslas is the peat bog of Cors Fochno. This is an area of poor drainage where soil is waterlogged. No trees grow on Cors Fochno. The climax community is bog, which is permanently dominated by the moss, *Sphagnum*.

(i) Explain why bog is described as the climax community. [1]

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(ii) Explain why trees are unable to survive in waterlogged soils. [2]

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The photograph shows a plant found on the bog called sundew (*Drosera rotundifolia*). This is an insectivorous plant which captures insects and digests them on its leaves.



The following equation shows the fixation of a nitrogen molecule ( $\text{N}_2$ ) to ammonia ( $\text{NH}_3$ ).



(d) Suggest why insectivorous plants such as sundew rely on catching prey. [3]

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